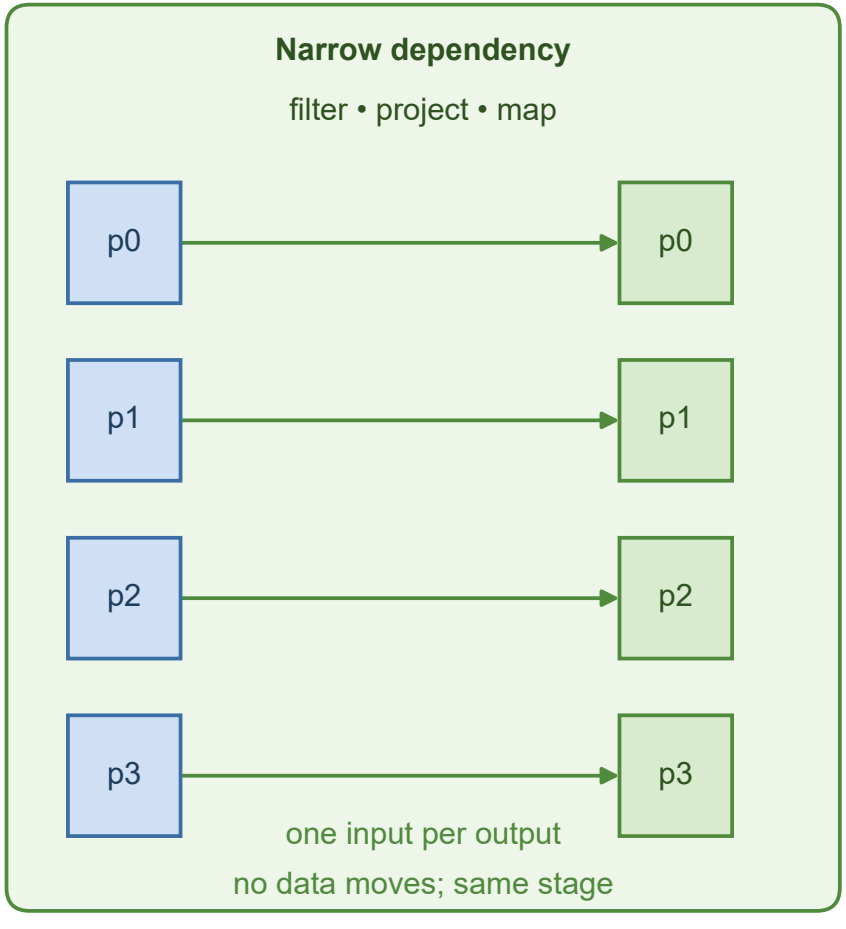
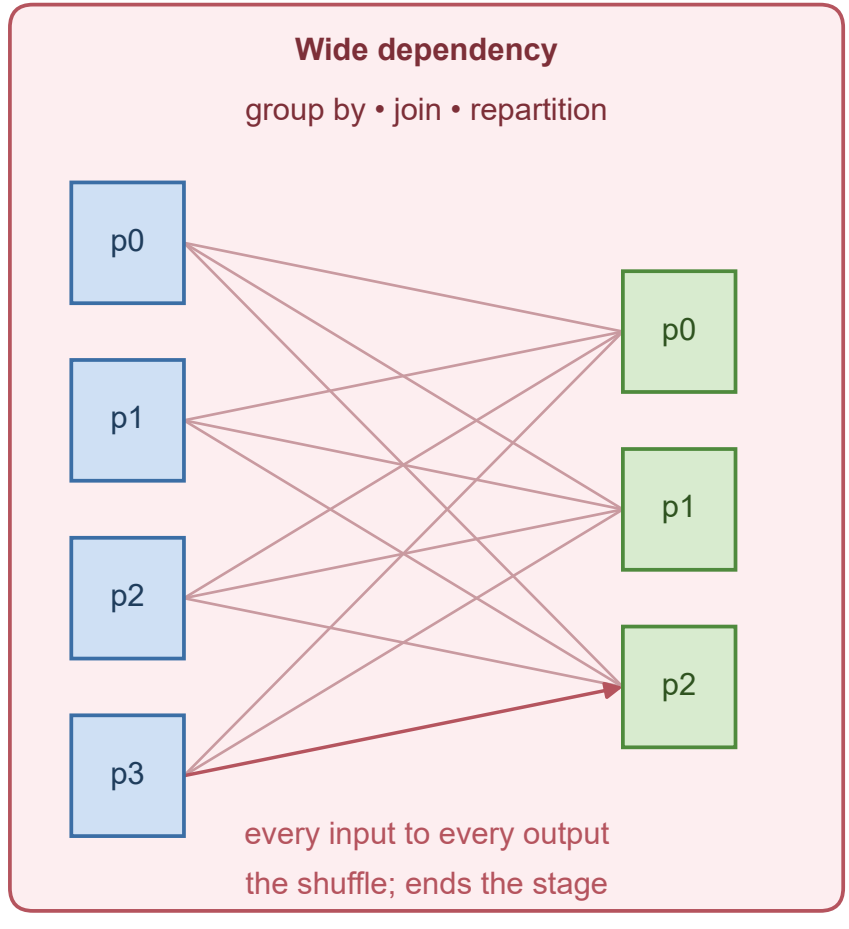

Narrow dependency
filter • project • map
p0
p1
p2
p3
p0
p1
p2
p3
one input per output
no data moves; same stage


Wide dependency
group by • join • repartition
p0
p1
p2
p3
p0
p1
p2
every input to every output
the shuffle; ends the stage